

SSTM PyPI Package for Optical Flow Estimation

Fisseha Ferede

July 10, 2026

Introduction

- ▶ **Goal:** Provide an easy-to-use PyPI package for dense optical flow estimation from image sequences.
- ▶ **Method:** SSTM, a deep-learning based spatiotemporal recurrent transformer model for multi-frame optical flow estimation.
- ▶ The package supports both two-frame and three-frame optical flow inference.

Installing SSTM PyPI Package

Installation

- ▶ First create and activate a clean Conda environment.
- ▶ Then install PyTorch and the SSTM PyPI package.

```
conda create -n sstm_env python=3.10  
conda activate sstm_env
```

```
pip install torch torchvision  
pip install sstm-flow
```

Importing SSTM

- ▶ After installation, SSTM can be imported directly from Python.
- ▶ The main function for simple inference is `estimate_flow`.

```
from sstm import estimate_flow
```

Running Optical Flow Inference

Two-frame Optical Flow

- ▶ For two-frame inference, SSTM estimates one optical flow field.
- ▶ The output **flow** represents motion from frame 1 to frame 2.

```
from sstm import estimate_flow

flow = estimate_flow(
    "checkpoints/ft-sintel.pth",
    "frame_0013.png",
    "frame_0014.png"
)
```

Three-frame Optical Flow

- ▶ For three-frame inference, SSTM estimates two flow fields.
- ▶ **f12**: optical flow from frame 1 to frame 2.
- ▶ **f23**: optical flow from frame 2 to frame 3.

```
from sstm import estimate_flow

f12, f23 = estimate_flow(
    "checkpoints/ft-sintel.pth",
    "frame_0013.png",
    "frame_0014.png",
    "frame_0015.png"
)
```


Using Already Loaded Images

- ▶ The package also supports images already loaded using OpenCV or NumPy.
- ▶ This is useful when images are preprocessed before flow estimation.

```
import cv2
from sstm import estimate_flow

img1 = cv2.imread("frame_0013.png")
img2 = cv2.imread("frame_0014.png")

flow = estimate_flow(
    "checkpoints/ft-sintel.pth",
    img1,
    img2
)
```

Recommended Inference Workflow

Faster Inference

- ▶ If running inference on many frame pairs, loading the model every time is inefficient.
- ▶ Instead, load the model once and reuse it for multiple image pairs.

```
import cv2
from sstm import load_model, infer_flow_pair

model = load_model("checkpoints/ft-sintel.pth")

img1 = cv2.imread("frame_0013.png")
img2 = cv2.imread("frame_0014.png")

flow = infer_flow_pair(model, img1, img2)
```

Saving Flow Outputs

- ▶ SSTM can save the predicted optical flow in **.flo** format.
- ▶ It can also save a standard color-wheel visualization of the flow field.

```
from sstm import estimate_flow

estimate_flow(
    "checkpoints/ft-sintel.pth",
    "frame_0013.png",
    "frame_0014.png",
    output_dir="results",
    save_flo=True,
    save_vis_outputs=True
)
```

Expected Output Files

- ▶ If `save_flo=True`, the flow field is saved as a `.flo` file.
- ▶ If `save_vis_outputs=True`, the flow visualization is saved as a PNG image.

```
results/  
|-- flow0001.flo  
|-- flow0001.png
```

Optical Flow Visualization

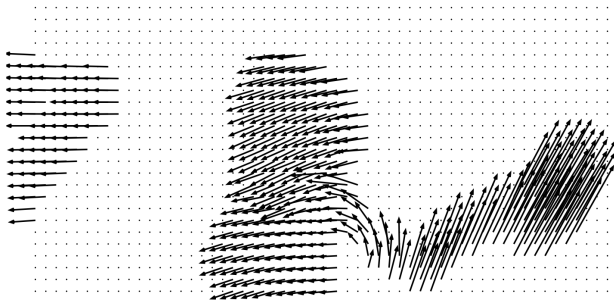
- ▶ Reference frame, target frame, and optical flow quiver plot between the two images.



(a) Reference frame



(b) Target frame



(c) Optical flow quiver plot

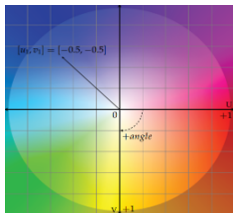
Optical Flow Visualization



(a) Reference frame



(b) Target frame



(c) Optical flow color wheel



(d) Optical flow color coding

Choosing the Right Checkpoint

Model Weight Recommendation

- ▶ **ft-speckle-sintel**: Recommended for local spatial deformation, biomechanics, elastography, biological tissue motion, and speckle tracking.
- ▶ **ft-sintel**: General-purpose model for non-rigid motion, natural scenes, animation, and research applications.
- ▶ **ft-kitti**: Recommended for rigid scene motion, autonomous driving, vehicle motion, and rigid object tracking.

Minimal Complete Example

```
from sstm import estimate_flow

flow = estimate_flow(
    "checkpoints/ft-speckle-sintel.pth",
    "img1.png",
    "img2.png",
    output_dir="results",
    save_flow=True,
    save_vis_outputs=True
)
```

References

- ▶ SSTM PyPI package page
- ▶ SSTM GitHub repository
- ▶ Ferede, F. A. and Balasubramanian, M. “SSTM: Spatiotemporal recurrent transformers for multi-frame optical flow estimation.” *Neurocomputing*, 2023.